



**CURIOSITY
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**A BETTER
WORLD**

TOPIC 2 – ATOMIC STRUCTURE AND THE PERIODIC TABLE

QEng Prep - Chemistry



FACULTY OF
ENGINEERING AND
APPLIED SCIENCE

KEY CONCEPTS

The key concepts covered in this lecture include:

- Explaining the relationship between the position of an element in the periodic table, its properties, and its electron configuration; and
- Identifying the characteristic properties of elements in each of the s, p, and d blocks of the periodic table.

INTRODUCTION

By convention, elements are organized in the Periodic Table, a structure that captures important patterns in their behavior.

Devised by Russian chemist Dmitri Mendeleev (1834–1907) in 1869, the table places elements into:

1. Columns (or **groups**)
2. Rows (or **periods**)

that share certain properties.

Periodic Table of the Elements

Period

Group

1 2 13 14 15 16 17 18

1 H 1.008 hydrogen 2 He 4.003 helium

2 Li 6.94 lithium 3 B 10.81 boron 4 Be 9.012 beryllium 5 C 12.01 carbon 6 N 14.01 nitrogen 7 O 16.00 oxygen 8 F 19.00 fluorine 9 Ne 20.18 neon

3 Na 22.99 sodium 10 Mg 24.31 magnesium 11 Al 26.98 aluminum 12 Si 28.09 silicon 13 P 30.97 phosphorus 14 S 32.06 sulfur 15 Cl 35.45 chlorine 16 Ar 39.95 argon

4 K 39.10 potassium 17 Ca 40.08 calcium 18 Sc 44.96 scandium 19 Ti 47.87 titanium 20 V 50.94 vanadium 21 Cr 52.00 chromium 22 Mn 54.94 manganese 23 Fe 55.85 iron 24 Co 58.93 cobalt 25 Ni 58.69 nickel 26 Cu 63.55 copper 27 Zn 65.38 zinc 28 Ga 69.72 gallium 29 Ge 72.63 germanium 30 As 74.92 arsenic 31 Se 78.97 selenium 32 Br 79.90 bromine 33 Kr 83.80 krypton

5 Rb 85.47 rubidium 34 Sr 87.62 strontium 35 Y 88.91 yttrium 36 Zr 91.22 zirconium 37 Nb 92.91 niobium 38 Mo 95.95 molybdenum 39 Tc 97.91 technetium 40 Ru 101.1 ruthenium 41 Rh 102.9 rhodium 42 Pd 106.4 palladium 43 Ag 107.9 silver 44 Cd 112.4 cadmium 45 In 114.8 indium 46 Sn 118.7 tin 47 Sb 121.8 antimony 48 Te 127.6 tellurium 49 I 126.9 iodine 50 Xe 131.3 xenon

6 Cs 132.9 cesium 51 Ba 137.3 barium 52 La-Lu 57-71 53 Hf 178.5 hafnium 54 Ta 180.9 tantalum 55 W 183.8 tungsten 56 Re 186.2 rhenium 57 Os 190.2 osmium 58 Ir 192.2 iridium 59 Pt 195.1 platinum 60 Au 197.0 gold 61 Hg 200.6 mercury 62 Tl 204.4 thallium 63 Pb 207.2 lead 64 Bi 209.0 bismuth 65 Po 209 [209] polonium 66 At 210 [210] astatine 67 Rn 222 [222] radon

7 Fr 223 [223] francium 68 Ra 226 [226] radium 69 Ac-Lr 89-103 70 Rf 261 [261] rutherfordium 71 Db 262 [262] dubnium 72 Sg 266 [266] seaborgium 73 Bh 264 [264] bohrium 74 Hs 277 [277] hassium 75 Mt 268 [268] meitnerium 76 Ds 271 [271] darmstadtium 77 Rg 272 [272] roentgenium 78 Cn 285 [285] copernicium 79 Uut 288 [288] ununtrium 80 Fl 289 [289] flerovium 81 Uup 289 [289] ununpentium 82 Lv 293 [293] livermorium 83 Uus 294 [294] ununseptium 84 Uuo 294 [294] ununoctium

* 57 La 138.9 lanthanum 58 Ce 140.1 cerium 59 Pr 140.9 praseodymium 60 Nd 144.2 neodymium 61 Pm 145 [145] promethium 62 Sm 150.4 samarium 63 Eu 152.0 europium 64 Gd 157.3 gadolinium 65 Tb 158.9 terbium 66 Dy 162.5 dysprosium 67 Ho 164.9 holmium 68 Er 167.3 erbium 69 Tm 168.9 thulium 70 Yb 173.1 ytterbium 71 Lu 175.0 lutetium

** 89 Ac 227 [227] actinium 90 Th 232.0 thorium 91 Pa 231.0 protactinium 92 U 238.0 uranium 93 Np 237 [237] neptunium 94 Pu 244 [244] plutonium 95 Am 243 [243] americium 96 Cm 247 [247] curium 97 Bk 247 [247] berkelium 98 Cf 251 [251] californium 99 Es 252 [252] einsteinium 100 Fm 257 [257] fermium 101 Md 258 [258] mendelevium 102 No 259 [259] nobelium 103 Lr 262 [262] lawrencium

Atomic number → 1

Symbol → H

Atomic mass → 1.008

Name → hydrogen

Color Code

Metal	Solid
Metalloid	Liquid
Nonmetal	Gas

THE PERIODIC TABLE

An element's position in the Periodic Table helps us to determine:

1. The element's physical state at room temperature – i.e. whether it is a solid, liquid, or gas.
2. The element's electron configuration – i.e. how the electrons are organized around the nucleus.

Knowing an element's electron configuration is often very important to us because it allows us to predict its chemical reactivity – i.e. how likely it is to form chemical bonds with atoms of other elements, and what these other elements will be.

Differences in the chemical reactivity of elements is based on the number and spatial distribution of their electrons. If two atoms have complementary electron patterns, it is very likely that they will react with one another to form a chemical bond. This is how molecules or compounds are created.

ELECTRON SHELLS AND THE BOHR MODEL

- An early model of the atom was developed in 1913 by the Danish scientist Niels Bohr (1885–1962).
- The Bohr model shows the atom as a central nucleus containing protons and neutrons, with the electrons in circular electron shells at specific distances from the nucleus, similar to planets orbiting around the sun.
- Each electron shell has a different energy level, with those shells closest to the nucleus being lower in energy than those farther from the nucleus.
- By convention, each shell is assigned a number and the symbol n .
- In order to move between shells, an electron must absorb or release an amount of energy corresponding exactly to the difference in energy between the shells.

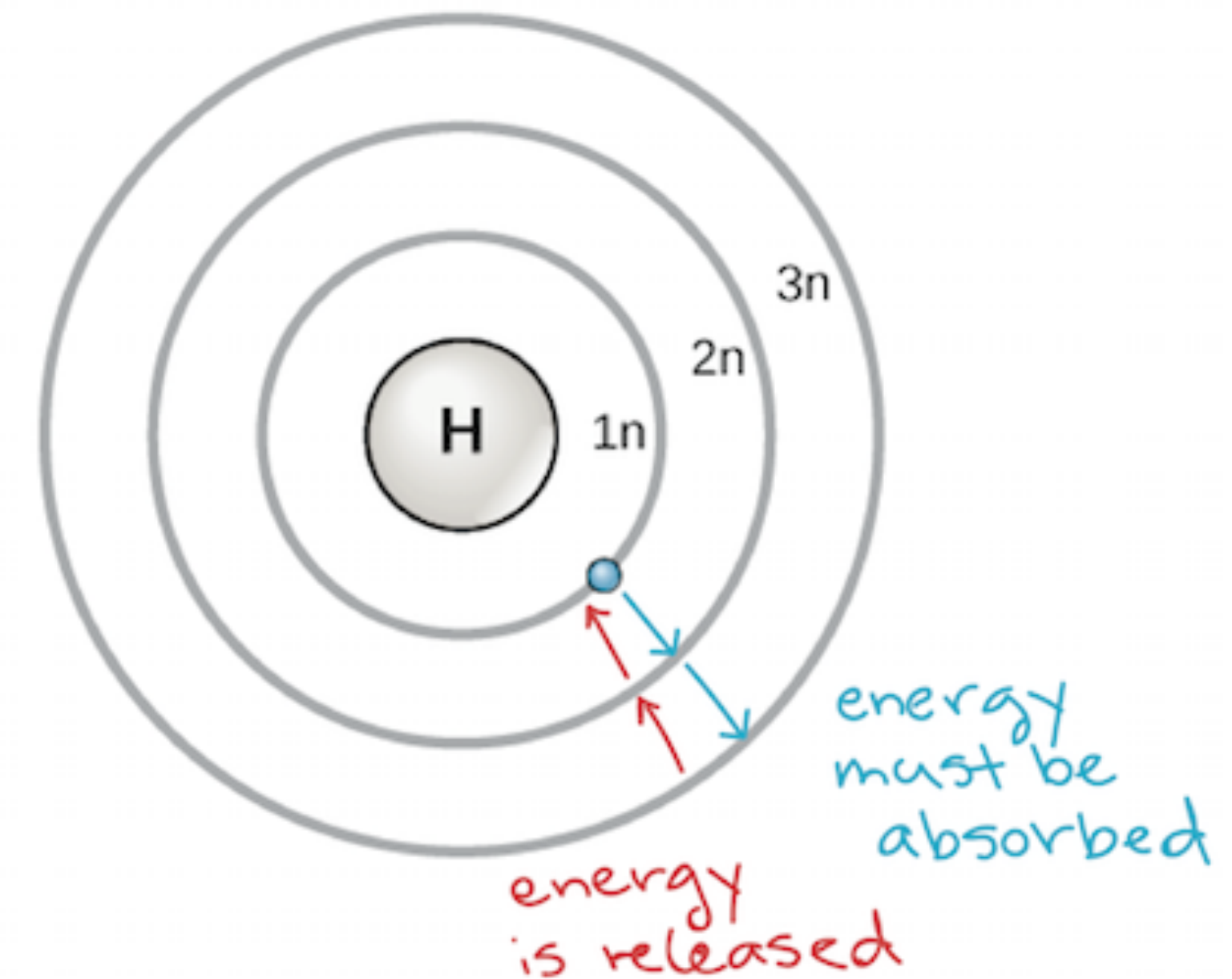


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ELECTRON SHELLS AND THE BOHR MODEL

General Ideas and Rules

- Atoms tend to take on the lowest-energy state possible, which is the most stable configuration. Electron shells of an atom are populated from the inside out - electrons fill up low-energy shells closer to the nucleus before moving into higher-energy shells further out.
 - Each shell can hold a specific number of electrons:
 1. **1n shell** – holds up to **2** electrons
 2. **2n shell** – holds up to **8** electrons
 3. **3n shell** – holds up to **18** electrons
 - The number of electrons in the outermost shell of a particular atom determines its reactivity. This outermost shell is known as the **valence shell**, and the electrons found in it are called **valence electrons**.
 - In general, atoms are most stable (least reactive) when their outermost electron shell is full. Most of the elements need eight electrons in their outermost shell in order to be stable. This rule of thumb is known as the **octet rule**.
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ELECTRON SHELLS AND THE BOHR MODEL

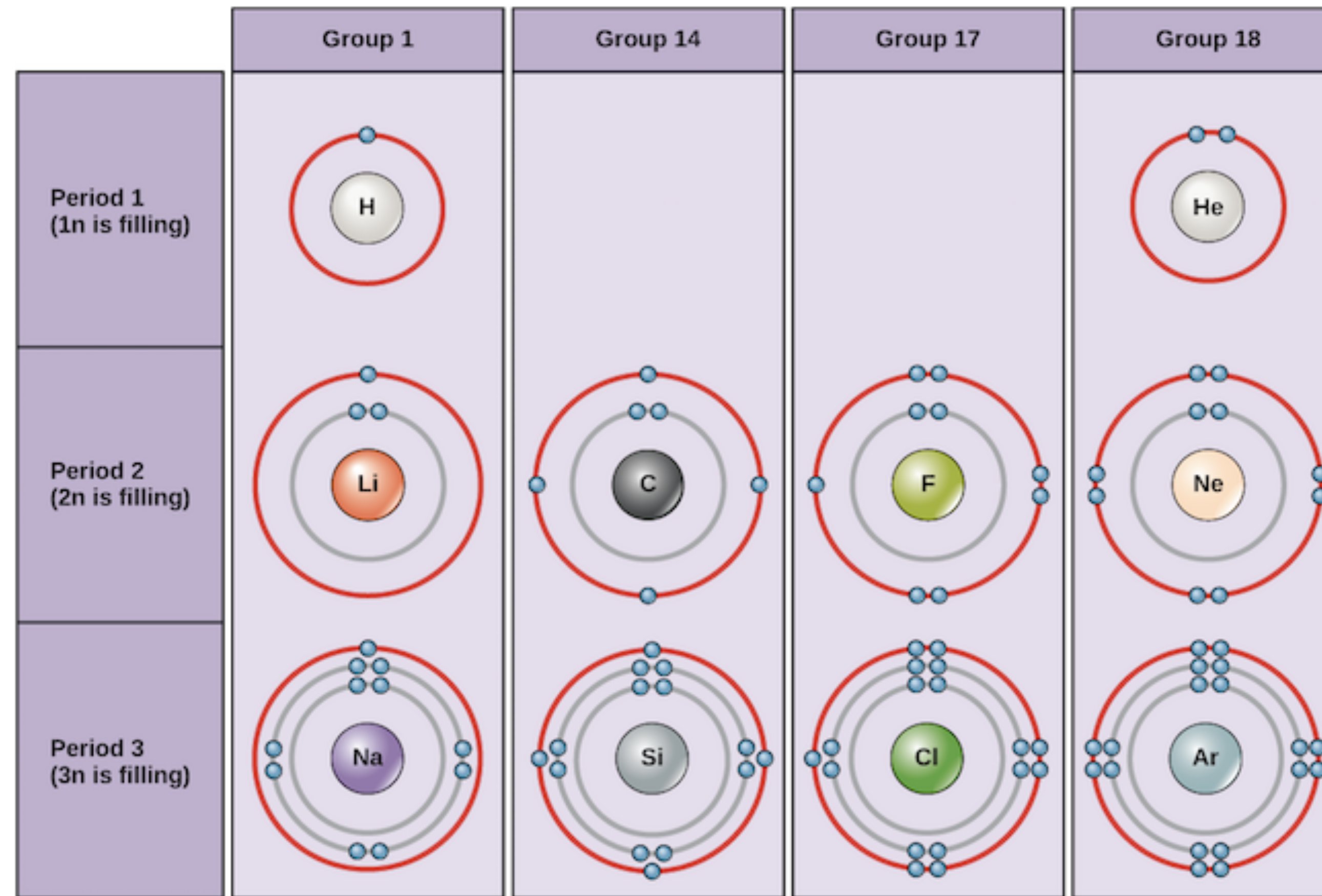


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ELECTRON CONFIGURATIONS AND THE PERIODIC TABLE

The periodic table is organized in such a way that an element's electron configuration is directly related to its position on the periodic table. There are three important trends we follow:

- 1. Atomic Number** - corresponds to the number of protons that an atom of that element has. In a neutral atom, this is also equal to the number of electrons.
- 2. Row (Period)** - corresponds to the filling of a different electron shell (for the first three rows). First-row elements (helium and hydrogen) place their electrons in the 1n shell, second-row elements start filling the 2n shell, and third-row elements continue with the 3n shell.
- 3. Column (Group)** - gives information about its number of valence electrons and reactivity

Periodic Table of the Elements

Period	Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1		1 H 1.008 hydrogen																	2 He 4.003 helium
2		3 Li 6.94 lithium	4 Be 9.012 beryllium											5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon
3		11 Na 22.99 sodium	12 Mg 24.31 magnesium											13 Al 26.98 aluminum	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.06 sulfur	17 Cl 35.45 chlorine	18 Ar 39.95 argon
4		19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.63 germanium	33 As 74.92 arsenic	34 Se 78.97 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton
5		37 Rb 85.47 rubidium	38 Sr 87.62 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.95 molybdenum	43 Tc [97] technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon
6		55 Cs 132.9 cesium	56 Ba 137.3 barium	57-71 La-Lu * lanthanum series	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.8 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po [209] polonium	85 At [210] astatine	86 Rn [222] radon
7		87 Fr [223] francium	88 Ra [226] radium	89-103 Ac-Lr ** actinide series	104 Rf [267] rutherfordium	105 Db [270] dubnium	106 Sg [271] seaborgium	107 Bh [270] bohrium	108 Hs [277] hassium	109 Mt [276] meitnerium	110 Ds [281] darmstadtium	111 Rg [282] roentgenium	112 Cn [285] copernicium	113 Uut [285] ununtrium	114 Fl [289] flerovium	115 Uup [288] ununpentium	116 Lv [293] livermorium	117 Uus [294] ununseptium	118 Uuo [294] ununoctium

* Lanthanide series (57-71): La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu

** Actinide series (89-103): Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr

Atomic number	1		
	H	Symbol	
	1.008	Atomic mass	
Name	hydrogen		

Color Code	
Metal	Solid
Metalloid	Liquid
Nonmetal	Gas

ELECTRON CONFIGURATIONS AND THE PERIODIC TABLE

- In general, the number of valence electrons is the same within a column and increases from left to right within a row. This means that group 1 elements have just one valence electron and group 18 elements have eight (with the exception of helium which only has two electrons total). This makes group number is a good predictor of how reactive each element will be:
 - **Group 18:** Helium, neon and argon have outer electron shells that are full or satisfy the octet rule. This makes them highly stable as single atoms. Because of their non-reactivity, they are called the inert gases or noble gases
 - **Group 1:** Hydrogen, lithium and sodium have just one electron in their outermost shells. They are unstable as single atoms, but can become stable by losing or sharing their one valence electron to become positively charged cations.
 - **Group 17:** Fluorine and chlorine, as Group 17 elements, have seven electrons in their outermost shells. They tend to achieve a stable octet by taking an electron from other atoms to become negatively charged anions.
 - **Group 14:** Carbon has four electrons in its outer shell. Carbon typically shares electrons to achieve a complete valence shell, forming bonds with multiple other atoms.

The diagram illustrates the placement of the lanthanide and actinide series within the periodic table. The lanthanide series, marked with an asterisk (*), consists of elements Ce through Lu and is associated with the 4f orbital. The actinide series, marked with a double asterisk (**), consists of elements Th through Lr and is associated with the 5f orbital. Both series are shown as separate rows below the main periodic table grid, with arrows indicating their insertion point between groups 2 and 3.

BLOCKS OF THE PERIODIC TABLE

s block

The s-block in the Periodic Table of elements occupies the **alkali metals** and **alkaline earth metals** (also known as Groups 1 and 2 respectively). Helium is also part of the s block.

Group 1:

- **Alkali metals** are among the most reactive metals.
- Tend to donate their electrons in reactions and have an oxidation state of +1.
- Lithium, sodium, and potassium float on water because of their low density.

Group 2:

- The heavier group 2 **alkali earth metals** are almost as reactive as the group 1 alkali metals.
- All elements in group 2 have two electrons in their valence shells. These are easily lost to gain stability, giving them an oxidation state of +2.

Period	Group 1	Group 2	18
1	H ¹ 1s		He ¹ 1s
2	Li ¹ 2s	Be ²	
3	Na ¹ 3s	Mg ²	
4	K ¹ 4s	Ca ²	
5	Rb ¹ 5s	Sr ²	
6	Cs ¹ 6s	Ba ²	
7	Fr ¹ 7s	Ra ²	

BLOCKS OF THE PERIODIC TABLE

p block

The p-block contains Groups 13, 14, 15, 16, 17, and 18, with the exception of helium (which is part of the s block). There is a maximum of six electrons that can occupy the p orbital which results in a maximum of 8 electrons in the 2n shell and 18 electrons in the 3n shell.

Group	Classification	State at Room Temperature	Valence Electrons	Reactivity
Group 13 (aka. the boron group)	metalloid – boron metals – aluminium, gallium, indium, thallium	all solids	3	fairly reactive
Group 14 (aka. the carbon group)	nonmetal – carbon metalloids – silicon, germanium metals – tin and lead	all solids	4	generally non-reactive
Group 15 (aka. the nitrogen group)	nonmetal – nitrogen, phosphorus metalloids – arsenic, antimony	all solids except for nitrogen (which is a gas)	5	reactivity varies with element

13	14	15
B 1	C 2	N 3
←		→ 2
Al 1	Si 2	P 3
←		→ 3
Ga 1	Ge 2	As 3
←		→ 4
In 1	Sn 2	Sb 3
←		→ 5
Tl 1	Pb 2	Bi 3
←		→ 6

BLOCKS OF THE PERIODIC TABLE

p block

The p-block contains Groups 13, 14, 15, 16, 17, and 18, with the exception of helium (which is part of the s block). There is a maximum of six electrons that can occupy the p orbital which results in a maximum of 8 electrons in the 2n shell and 18 electrons in the 3n shell.

Group	Classification	State at Room Temperature	Valence Electrons	Reactivity
Group 16 (aka. the oxygen group)	nonmetals – oxygen, sulfur, selenium metalloid - tellurium metal – polonium	all solids except for oxygen (which is a gas)	6	very reactive
Group 17 (aka. the halogens)	all nonmetals	gases – fluorine, chlorine liquid – bromine solids – iodine and astatine	7 (in 2n and 3n)	react readily with metals to form salts
Group 18 (aka. the noble gases)	all nonmetals	all monatomic gases	8 (in 2n and 3n)	unreactive/ inert

16	17	18
O 4	F 5	Ne 6
p →		
S 4	Cl 5	Ar 6
p →		
Se 4	Br 5	Kr 6
p →		
Te 4	I 5	Xe 6
p →		
Po 4	At 5	Rn 6
p →		

BLOCKS OF THE PERIODIC TABLE

d block

- The d-block elements are found in Groups 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 of the Periodic Table.
- d block elements are also known as the transition metals.
- The transition metals are generally characterized by partially filled d subshells in the free elements or their cations.
- Transition metals demonstrate a wide range of chemical behaviors: some transition metals are strong reducing agents, whereas others have very low reactivity.

3	4	5	6	7	8	9	10	11	12
Sc ¹	Ti ²	V ³	Cr ⁴	Mn ⁵	Fe ⁶	Co ⁷	Ni ⁸	Cu ⁹	Zn ¹⁰
←					3d				→
Y ¹	Zr ²	Nb ³	Mo ⁴	Tc ⁵	Ru ⁶	Rh ⁷	Pd ⁸	Ag ⁹	Cd ¹⁰
←					4d				→
La ^{*1}	Hf ²	Ta ³	W ⁴	Re ⁵	Os ⁶	Ir ⁷	Pt ⁸	Au ⁹	Hg ¹⁰
←					5d				→
Ac ^{**1}	Rf ²	Db ³	Sg ⁴	Bh ⁵	Hs ⁶	Mt	Ds	Rg	Cn
←					6d				→